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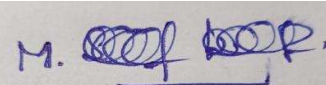
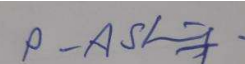
THE PATENT RULES,
2003

COMPLETE SPECIFICATION

(See Section 10 and Rule
13)

Title of Invention

A system and method of Machine Learning Based Personalized Learning Device

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TECHNICAL FIELD

The present invention relates generally to the field of educational technology, artificial intelligence, and machine learning systems. More particularly, the invention relates to a Machine Learning Based Personalized Learning Device and method configured to provide adaptive, intelligent, and customized learning experiences for users based on their learning behavior, performance, preferences, and cognitive abilities.

The invention integrates machine learning algorithms, data analytics, user profiling techniques, and intelligent recommendation systems to improve learning efficiency, student engagement, knowledge retention, and educational outcomes. The system is capable of collecting and analyzing educational data from multiple learning activities including quizzes, assignments, online classes, assessments, interaction patterns, attendance records, and subject-specific performance metrics.

The disclosed invention may be implemented in educational institutions, online learning platforms, training centers, corporate learning systems, remote education environments, and self-learning applications. The device supports real-time adaptive learning by dynamically modifying educational content, difficulty levels, learning schedules, and teaching methodologies according to individual learner requirements.

The invention further relates to predictive analytics systems capable of identifying learning gaps, predicting academic performance, and recommending personalized study materials and improvement strategies. The system may utilize supervised learning, unsupervised learning, reinforcement learning, deep learning, natural language processing, and recommendation algorithms to continuously optimize learning pathways.

Additionally, the invention supports integration with cloud computing infrastructure, mobile devices, wearable educational devices, smart classrooms, virtual learning environments, and Internet of Things (IoT)-enabled educational systems. The invention therefore provides an intelligent educational framework for automated personalized learning, adaptive content delivery, academic performance enhancement, and efficient knowledge management using machine learning technologies.

BACKGROUND

Traditional educational systems generally follow a standardized teaching methodology in which the same content, pace, and instructional approach are applied uniformly to all learners regardless of their individual abilities, learning speed, interests, or cognitive differences. Such conventional systems often fail to address personalized educational requirements, resulting in reduced student engagement, ineffective learning experiences, lower knowledge retention, and inconsistent academic performance.

With the rapid growth of digital education platforms and online learning systems, large volumes of educational data are being generated through student interactions, assignments, examinations, attendance records, and online activities. However, existing educational platforms frequently lack intelligent mechanisms for effectively analyzing such data to provide customized learning experiences tailored to individual users.

Current adaptive learning systems are often limited in functionality and may rely on predefined rules rather than intelligent machine learning models. These systems may not accurately identify student weaknesses, predict future learning outcomes, or dynamically adjust educational content in real time. Furthermore, many existing platforms provide generic recommendations that do not sufficiently consider learner behavior, emotional engagement, subject difficulty levels, or knowledge progression patterns.

In many educational environments, teachers face challenges in monitoring the progress of a large number of students simultaneously. Manual evaluation methods are time-consuming and may not provide real-time insights into individual learner performance. Similarly, students often struggle to identify suitable study materials, optimal learning paths, and effective revision strategies.

The increasing adoption of remote learning and digital classrooms has further highlighted the need for intelligent educational systems capable of delivering adaptive, scalable, and automated learning support. There exists a requirement for a system that can continuously analyze user learning patterns, detect knowledge gaps, recommend personalized educational content, and provide predictive academic insights.

Accordingly, there is a need for an advanced Machine Learning Based Personalized Learning Device capable of utilizing artificial intelligence and data analytics techniques to create

customized learning experiences, improve academic outcomes, automate learner assessment, and enhance overall educational efficiency across multiple learning environments.

SUMMARY OF THE INVENTION

The present invention provides a Machine Learning Based Personalized Learning Device and method configured to deliver intelligent, adaptive, and customized educational experiences using machine learning and artificial intelligence technologies.

In one embodiment, the system comprises a data acquisition module, learner profiling engine, preprocessing module, machine learning analytics engine, recommendation engine, adaptive content delivery module, performance monitoring system, and visualization dashboard. These components communicate through secure processing architecture and may operate on cloud-based, on-premises, or hybrid educational infrastructures.

The data acquisition module collects educational data from multiple sources including online learning platforms, quizzes, examinations, assignments, attendance systems, classroom interactions, and user behavioral activities. The collected data may include subject performance records, response times, engagement metrics, learning preferences, and progress indicators.

The preprocessing module performs data cleaning, normalization, feature extraction, and transformation operations to convert raw educational data into machine-readable formats suitable for machine learning analysis.

The learner profiling engine creates individualized learner profiles based on academic history, cognitive behavior, subject preferences, learning speed, and interaction patterns. The profiles are continuously updated as new learning data becomes available.

The machine learning analytics engine utilizes supervised learning, clustering algorithms, neural networks, reinforcement learning, and predictive analytics models to identify learning gaps, predict academic performance, classify learner behavior, and optimize personalized study paths.

The recommendation engine generates customized educational recommendations including

study materials, practice questions, revision schedules, learning videos, and subject-specific exercises according to learner requirements. The adaptive content delivery module dynamically adjusts content complexity, presentation methods, and learning pace in real time.

The invention may further include predictive performance analysis for identifying at-risk learners and generating automated intervention strategies. Real-time dashboards provide educators, students, and administrators with detailed learning analytics, progress reports, and performance insights.

The disclosed invention improves learning efficiency, academic engagement, personalized education delivery, and knowledge retention while reducing manual educational management efforts.

OBJECTIVE OF INVENTION

The primary objective of the present invention is to provide a Machine Learning Based Personalized Learning Device capable of delivering adaptive and customized learning experiences according to individual learner behavior, academic performance, and educational requirements.

Another objective of the invention is to utilize machine learning and artificial intelligence algorithms for analyzing educational data and identifying learner strengths, weaknesses, preferences, and knowledge gaps in real time.

A further objective of the invention is to provide an intelligent recommendation system capable of suggesting personalized study materials, assignments, learning schedules, revision plans, and educational resources for improving academic performance and knowledge retention.

Another objective of the invention is to enable dynamic content adaptation wherein the system automatically adjusts learning difficulty levels, teaching methods, and educational pathways based on learner progress and engagement.

A further objective of the invention is to improve student engagement and motivation through interactive and personalized educational experiences that suit individual learning capabilities and interests.

Another objective of the invention is to provide predictive analytics mechanisms capable of forecasting academic outcomes, identifying at-risk learners, and recommending corrective actions before performance deterioration occurs.

A further objective of the invention is to reduce the workload of educators by automating learner monitoring, assessment analysis, performance tracking, and personalized recommendation generation.

Another objective of the invention is to support scalable educational environments including schools, colleges, online learning platforms, corporate training systems, and remote education infrastructures.

A further objective of the invention is to ensure secure handling of educational data through encrypted communication, access control mechanisms, and privacy protection techniques.

Another objective of the invention is to provide continuous self-learning capabilities wherein machine learning models improve recommendation accuracy and learning optimization using newly collected educational data.

A further objective of the invention is to integrate with cloud computing systems, mobile applications, smart classrooms, wearable devices, and Internet of Things-enabled educational platforms for seamless learning accessibility.

Thus, the present invention aims to provide an intelligent, automated, scalable, and adaptive personalized learning framework that enhances educational quality, learner satisfaction, academic efficiency, and knowledge acquisition using machine learning technologies.

BRIEF DESCRIPTION OF DRAWINGS

Data Collection

The system collects educational data from multiple sources such as online classes, quizzes, assignments, examinations, attendance systems, mobile applications, and student interaction activities.

Data Preprocessing

The collected data is cleaned, filtered, normalized, and transformed into structured machine-readable formats. Duplicate and irrelevant information is removed for accurate analysis.

Learner Profile Generation

The system creates personalized learner profiles based on academic history, learning speed, subject preferences, behavior patterns, cognitive abilities, and engagement levels.

Machine Learning Analysis

Machine learning algorithms analyze learner data to identify strengths, weaknesses, learning gaps, performance trends, and preferred learning methods.

Personalized Recommendation System

Based on analytical results, the system recommends customized study materials, learning videos, practice exercises, revision schedules, and subject-specific content.

Adaptive Learning Mechanism

The device dynamically adjusts content difficulty level, learning pace, and teaching methods according to learner performance and progress.

Performance Monitoring

The system continuously tracks learner progress, assessment scores, attendance, completion rates, and engagement metrics in real time.

Predictive Analytics

The predictive module forecasts academic performance and identifies at-risk learners, enabling early intervention and improvement strategies.

Visualization and Reporting

A dashboard provides graphical reports, performance analytics, learning progress charts, and educational insights for students, teachers, and administrators.

Continuous Self-Learning

The machine learning models continuously retrain using newly collected educational data to improve recommendation accuracy and learning optimization over time.

DETAILED DESCRIPTION

The present invention relates to a Machine Learning Based Personalized Learning Device and method configured to provide intelligent, adaptive, and customized educational experiences for users through artificial intelligence and machine learning technologies. The invention enables automated learner analysis, adaptive content delivery, personalized recommendations, academic performance prediction, and real-time educational monitoring.

The disclosed system comprises a data acquisition module, preprocessing engine, learner profiling unit, machine learning analytics engine, recommendation engine, adaptive learning module, performance monitoring system, communication interface, and visualization dashboard. These components collectively operate to analyze learner behavior and optimize educational content according to individual learning requirements.

The data acquisition module is configured to collect educational data from multiple learning sources including online learning platforms, educational databases, mobile applications, classroom systems, quizzes, examinations, assignments, attendance records, interactive sessions, and user behavioral activities. The collected data may include academic scores, response time, subject preferences, learning frequency, engagement metrics, browsing patterns, and content interaction history.

In one embodiment, the system supports real-time and batch-mode data collection techniques

for continuous monitoring of learner activities. The data acquisition module may further communicate with cloud-based educational platforms, smart classroom devices, wearable learning devices, and Internet of Things (IoT)-enabled educational systems.

The preprocessing engine performs cleaning, normalization, filtering, transformation, feature extraction, and deduplication operations on the collected educational data. Since raw learning data may contain incomplete records, inconsistent formats, and irrelevant information, the preprocessing engine converts such data into structured machine-readable formats suitable for machine learning analysis.

The learner profiling unit generates individualized learner profiles based on historical academic records, behavioral patterns, cognitive abilities, preferred learning styles, subject performance, attention span, and interaction history. The learner profiles are dynamically updated as additional educational data becomes available. The profiling unit may classify learners into multiple categories such as beginner, intermediate, advanced, slow learner, fast learner, visual learner, or analytical learner.

The machine learning analytics engine forms the core component of the invention. The analytics engine utilizes machine learning algorithms including supervised learning, unsupervised learning, reinforcement learning, decision trees, clustering algorithms, neural networks, deep learning models, and natural language processing techniques for educational data analysis.

In one implementation, the analytics engine identifies learning gaps by comparing learner performance against predefined academic benchmarks and subject competency requirements. The system analyzes quiz scores, assessment patterns, response behavior, and learning progress to determine weak subject areas requiring additional educational support.

The recommendation engine generates personalized educational recommendations based on learner profiles and analytical results. The recommendations may include customized study materials, learning videos, practice exercises, revision plans, interactive tutorials, examination preparation strategies, and subject-specific assignments. The recommendation engine continuously improves recommendation accuracy using adaptive learning mechanisms and feedback analysis.

The adaptive learning module dynamically adjusts educational content according to learner

performance and engagement level. For example, if a learner demonstrates difficulty in understanding a particular topic, the system automatically provides simplified explanations, additional practice sessions, graphical representations, or alternative teaching methods. Similarly, advanced learners may receive higher-difficulty content and accelerated learning pathways.

The performance monitoring system continuously tracks learner progress and educational outcomes. The monitoring system evaluates metrics such as completion rates, assessment scores, learning speed, engagement duration, attendance, and conceptual understanding. Real-time monitoring enables early detection of academic decline or disengagement.

In another embodiment, the invention includes predictive analytics capabilities configured to forecast learner performance and identify at-risk students. The predictive system analyzes historical educational data and behavioral patterns to estimate future academic outcomes. Based on predictive results, the system may recommend intervention strategies such as additional tutoring, revised study schedules, targeted practice exercises, or instructor support.

The visualization dashboard provides graphical representations of educational analytics, learner progress reports, subject performance trends, attendance statistics, engagement metrics, and predictive academic insights. Teachers, administrators, parents, and learners may access the dashboard through web interfaces or mobile applications for monitoring educational activities in real time.

The invention further supports multilingual educational content and natural language processing-based interaction systems. Learners may communicate with the system through text-based queries, voice commands, or conversational educational assistants. The system may additionally provide automated doubt-solving assistance and intelligent question-answering functionalities.

The communication interface enables secure communication between system components and external educational platforms. The invention may utilize encrypted communication protocols, secure authentication mechanisms, and role-based access control systems to ensure confidentiality and integrity of educational data.

In one exemplary implementation, a student accesses the personalized learning device through a mobile application. The system collects data regarding completed assignments,

quiz scores, and time spent on learning modules. The machine learning engine identifies weak performance in mathematics and automatically recommends simplified tutorials, practice problems, and visual explanations. As the learner improves, the system gradually increases difficulty levels and updates the learner profile accordingly.

In another embodiment, educational institutions may deploy the system across multiple classrooms and online learning environments. Teachers may receive automated reports identifying students requiring additional academic assistance. Administrators may utilize analytical dashboards for monitoring institutional learning performance and curriculum effectiveness.

The invention offers several advantages over conventional educational systems. These advantages include personalized learning experiences, improved student engagement, enhanced knowledge retention, automated academic monitoring, reduced educator workload, real-time adaptive learning, predictive performance analysis, scalable deployment, and intelligent educational decision-making.

The disclosed invention may be implemented using cloud computing infrastructure, edge computing systems, embedded processors, mobile devices, desktop systems, or distributed computing environments. The software modules may be developed using programming languages and machine learning frameworks including Python, Java, TensorFlow, PyTorch, and Scikit-learn.

Thus, the present invention provides an intelligent Machine Learning Based Personalized Learning Device and method capable of delivering adaptive, scalable, and automated personalized education while improving academic efficiency, learner satisfaction, and educational outcomes through advanced machine learning technologies.

We CLAIMS:

A Machine Learning Based Personalized Learning Device comprising a data acquisition module, preprocessing engine, learner profiling unit, machine learning analytics engine, recommendation engine, adaptive learning module, performance monitoring system, and visualization dashboard configured to provide personalized educational experiences.

The device of claim 1, wherein the data acquisition module collects educational data from online learning platforms, examinations, assignments, classroom systems, mobile applications, and user interaction activities.

The device of claim 1, wherein the preprocessing engine performs data cleaning, normalization, feature extraction, filtering, transformation, and deduplication of educational data for machine learning analysis.

The device of claim 1, wherein the learner profiling unit generates dynamic learner profiles based on academic performance, behavioral patterns, cognitive abilities, learning preferences, and interaction history.

The device of claim 1, wherein the machine learning analytics engine utilizes supervised learning, unsupervised learning, deep learning, reinforcement learning, and predictive analytics algorithms to identify learning gaps and optimize educational pathways.

The device of claim 1, wherein the recommendation engine provides customized study materials, practice exercises, learning videos, revision schedules, and subject-specific educational content according to learner requirements.

The device of claim 1, wherein the adaptive learning module dynamically adjusts content difficulty level, learning pace, and teaching methodology based on learner performance and engagement.

A method for machine learning based personalized learning, comprising collecting learner data, preprocessing educational information, generating learner profiles, analyzing performance using machine learning algorithms, recommending personalized educational resources, and dynamically adapting learning content for improving educational outcomes.

ABSTRACT

The present invention relates to a Machine Learning Based Personalized Learning Device and method for delivering adaptive and customized educational experiences using artificial intelligence and machine learning technologies. The system comprises a data acquisition module, preprocessing engine, learner profiling unit, machine learning analytics engine, recommendation engine, adaptive learning module, performance monitoring system, and visualization dashboard. The device collects educational data from online learning platforms, assignments, examinations, classroom systems, and user interactions. Machine learning algorithms analyze learner behavior, academic performance, cognitive patterns, and engagement metrics to identify learning gaps and generate personalized educational recommendations. The adaptive learning module dynamically adjusts content difficulty, learning pace, and instructional methods according to individual learner requirements. The system further provides predictive academic analytics, real-time progress monitoring, automated performance evaluation, and intelligent study recommendations. The invention improves learner engagement, educational efficiency, knowledge retention, and academic outcomes while reducing manual educational management efforts through scalable and intelligent personalized learning mechanisms.

Figure

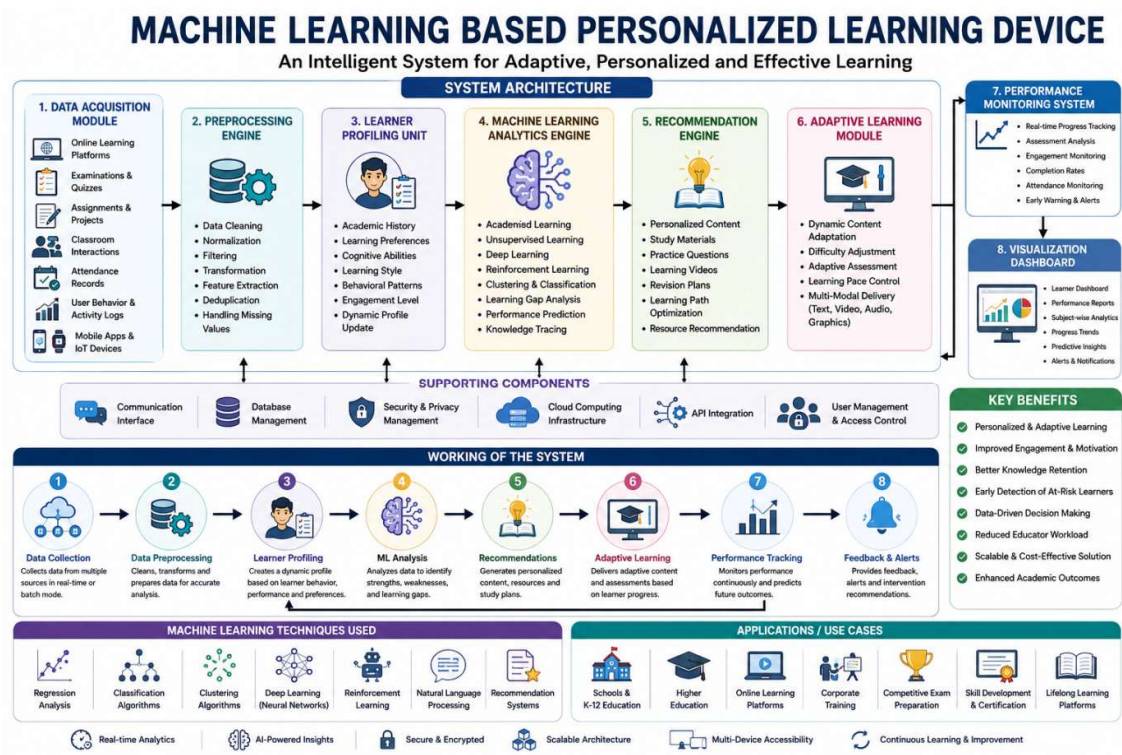


Figure 1: block diagram of work